

Abbreviations (hiabbrev.dtx)

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This package provides for abbreviating texts according to tables of abbreviations. It also provides functions for removing spaces between dotted initials. The abbreviation algorithms are fully expandable so that they may be used within `\edef` definitions.

```
\ProvidesPackage{hiabbrev}[2017/08/14 Abbreviations]
\RequirePackage{etoolbox}
\RequirePackage{histrings}
```

1 Abbreviation Classes

There may be multiple abbreviation classes, namely separate schemes of abbreviations with different words abbreviated in each.

An abbreviation class is defined in the macro `\abb@class@<name>`. Within that class are a further set of macros of the form `\abb@class@<name>@<word>`, where `<word>` is the first word of the text to be abbreviated. The value of the macro is a list of two-item pairs of the form:

```
{<remaining text>}{<abbrev.>}
```

The list is organized with the longest abbreviations first, ensuring that longer abbreviations take precedence over shorter ones.

`\abb@new` `{<class>}` creates a new abbreviation class.

```
\def\abb@new#1{\@namedef{abb@class@#1}{}}%
```

`\abb@add` `{<class>}{<word>}{<abbrev>}`

Adds an item to the abbreviation class. `<class>` is the class, `<word>` the unabbreviated form, and `{<abbrev>}` the abbreviated form.

If the unabbreviated form starts with an empty set of braces, then the given abbreviation pair is treated as an initial-text abbreviation, which will only abbreviate at the beginning of a text. That is, the abbreviation pairs:

```
{ }Building → Bldg.
Building → Build.
```

would produce abbreviations as follows:

The Building and Loan Office → The Build. and Loan Office
 Building and Loan Office → Bldg. and Loan Office

```
\def\abb@add#1#2#3{%
  \find@word{#2}{%
    \abb@add@punct{#1}{#3}%
  }{%
    \abb@add@group{#1}{#3}%
  }{%
    \abb@add@{#1}{#3}%
  }{%
    \abb@add@error{nothing}{}{}%
  }%
}
```

Processes an abbreviation pair where the unabbreviated text starts with a group. #1 is the class, #2 the abbreviated text, #3 the group contents, and #4 the remainder.

If the group contains any text, then this is an error. Otherwise, an empty group at the beginning indicates an abbreviation that only matches at the start of the text. The initial-text abbreviation is implemented by storing the abbreviation as a continuation of a zero-width word.

```
\def\abb@add@group#1#2#3#4{%
  \ifstrempy{#3}{%
    \abb@add@{#1}{#2}{#3}{#4}%
  }{%
    \abb@add@error{group}{#3}{#4}%
  }%
}
\def\abb@add@error#1#2#3{%
  \PackageError{abbrev}{%
    Abbreviation text `#2#3' should not start with #1 `#2'
  }{%
    Review the text to be abbreviated
  }%
}
\def\abb@add@punct#1#2#3#4{%
  \@test \ifcat \relax\noexpand#3\fi{%
    \abb@add@error{control sequence}{#3}{#4}%
  }{%
    \abb@add@{#1}{#2}{#3}{#4}%
  }%
}
```

#1 is the class, #2 is the abbreviated form, #3 the first word, #4 the remainder.

```
\def\abb@add@#1#2#3#4{%
  \make@find@start{#3}% I think this is superfluous
  \expandafter\abb@add@@
  \csname abb@class@#1@#3\endcsname
  {#2}{#4}%
}
```

#1 is the macro containing the class and first word, #2 the abbreviated form, #3 the remainder of the abbreviation.

```
\def\abb@add@@#1#2#3{%
  \make@find@start{#3}%
  \ifx#1\relax
    \def#1{{#3}{#2}}%
  \else
    \expandafter\abb@add@insert#1\@stop{#1}{#3}{#2}{}%
  \fi
}
```

Decides where to insert the new abbreviation. The key is that, because abbreviation matches are searched in sequence, longer strings need to come before substrings. For example, an abbreviation list:

{Aaa Bbb}{A.B.}{Aaa Bbb Ccc}{A.B.C.}

would be problematic since to abbreviate “Aaa Bbb Ccc Y”, the first abbreviation pair would match and the second would never be considered. So the list properly needs to be ordered:

{Aaa Bbb Ccc}{A.B.C.}{Aaa Bbb}{A.B.}

#1 and #2 are the abbreviation pair being considered, #3 is the rest of the list, #4 is the list macro, #5 the unabbreviated text to match, #6 the abbreviation, and #7 any material to prepend to the list.

```
\def\abb@add@insert#1#2#3\@stop#4#5#6#7{%
  \find@start{#1}{#5}{%
    \def#4{#7{#5}{#6}{#1}{#2}#3}%
    \@gobble
  }{%
    \ifstrempy{#3}{%
      \def#4{#7{#1}{#2}{#5}{#6}}%
    }{%
      \abb@add@insert#3\@stop{#4}{#5}{#6}{#7{#1}{#2}}%
    }%
  }%
}
```

2 Algorithm for Abbreviation

`\abb@abbrev` $\{\langle class \rangle\}\{\langle text \rangle\}\{\langle callback \rangle\}$ is the entrypoint for abbreviating a text. $\langle class \rangle$ is the class, $\langle text \rangle$ is the text to abbreviate, and $\langle callback \rangle$ is a callback function for the result.

Text is abbreviated by reading each token in the text, using `\find@word` from `histrings.dtx`.

1. If the token is neither a word nor punctuation, do not process the item and continue. In particular, text in groups is ignored for abbreviation purposes.
2. For words and regular punctuation, retrieve the list of abbreviations starting with this word. If there are none, then continue.
3. Iterate through each of the abbreviations associated with the word, and see if the remainder of the text to be abbreviated starts with any of the abbreviation texts. If so, then insert the relevant abbreviation text.
 - If the abbreviation fully deletes text, then check that the abbreviated word is surrounded by spaces (or is at the beginning or end of the text). If not, then the abbreviation is not performed since it would result in disconnected punctuation. If it is, then delete any excess space.
 - If the abbreviation ends with a dot or space, then ensure that insertion of the abbreviation does not create a double dot or space.

Throughout the below description, the “accumulated text” is the text that has been abbreviated at the beginning of the string, and the “remaining text” is the text that has not been processed for abbreviation at the end of the string. In the midst of determining whether to abbreviate something, the accumulated text will not include the proposed abbreviation.

To provide the special feature of initial-text abbreviations, we assume that the text starts off with a zero-width word, and try to abbreviate it. Since the initial-text abbreviations are stored under the command for zero-width words, this produces the desired result.

```
\def\abb@abbrev#1#2#3{%
  \abb@abbrev@wd{}{#2}{#1}{#3}%
}
```

Read the next token off of the remaining text to determine how to abbreviate it. #1 is the remaining text to abbreviate, #2 is the accumulated text, #3 is the class, #4 is the callback. (#2-#4 are left on the stack.)

```
\def\abb@abbrev@acc#1{%
  \find@word{#1}\abb@abbrev@punct\abb@abbrev@grp\abb@abbrev@wd\abb@abbrev@end
}
```

Given that the first token of the remaining text is punctuation, determine how to abbreviate the symbol. If the first token is a macro, nothing is done. Otherwise, it is abbreviated as if it were a word in itself.

#1 is the punctuation, #2 the remaining text, #3 the accumulated text, #4 the class, #5 the callback. (#4-#5 are left on the stack.)

```
\def\abb@abbrev@punct#1#2#3{%
  \@test \ifcat \relax\noexpand#1\fi{%
    \abb@abbrev@acc{#2}{#3#1}%
  }{%
    \abb@abbrev@wd{#1}{#2}{#3}%
  }%
}
```

Given that the token being abbreviated is a group, nothing is done. The content of the group is appended to the accumulated text, and the remaining text is further processed.

#1 is the group, #2 the remaining text, #3 the accumulated text, #4 the class, #5 the callback. (#4-#5 are left on the stack.)

```
\def\abb@abbrev@grp#1#2#3{%
  \abb@abbrev@acc{#2}{#3{#1}}%
}
```

Given that there is no more text left to abbreviate, invoke the callback with the accumulated text.

#1 is the accumulated text, #2 the class, #3 the callback.

```
\def\abb@abbrev@end#1#2#3{%
  #3{#1}%
}
```

Given that the token to be abbreviated is a word or abbreviatable punctuation, then search for an abbreviation starting with this word. If there is no defined abbreviation for the word, then continue to the next token. Otherwise, call `\abb@abbrev@wd@`, placing on the stack the list of abbreviations associated with the word.

#1 is the word under consideration, #2 the remaining text (not including the word under consideration), #3 the accumulated text, #4 the class, #5 the callback. (#5 is left on the stack.)

```
\def\abb@abbrev@wd#1#2#3#4{%
  \@ifundefined{abb@class@#4@#1}{%
    \abb@abbrev@acc{#2}{#3#1}{#4}%
  }{%
    \expandafter\expandafter\expandafter\abb@abbrev@wd@
      \csname abb@class@#4@#1\endcsname \@stop
    {#1}{#2}{#3}{#4}%
  }%
}
```

Given that the word under consideration has abbreviations, consider all possible abbreviations starting with this word to see if they match the remaining text.

#1 is a candidate abbreviatable text; #2 is the candidate abbreviation; #3 is the remainder of that list. #4-#8 are the args to `\abb@abbrev@wd`, namely #4 the word under consideration, #5 the remaining text not including the word under consideration, #6 the accumulated text, #7 the class, and #8 the callback. (#7-#8 are left on the stack.)

```
\def\abb@abbrev@wd@#1#2#3\@stop#4#5#6{%
  \find@start{#1}{#5}{%
    \abb@abbrev@found{#6}{#2}%
  }{%
    \@iden
  }%
}
```

This is the “bailout” content, to be executed if #1 does not match the remaining text. It is executed if the `\find@start` above fails to match, or it is presented as argument #4 to `\abb@abbrev@found`. What it does is:

- If there are no remaining abbreviation candidates, then attach the word under consideration to the accumulated text and proceed to abbreviate the remaining text.

- If there are remaining abbreviation candidates, then proceed to the next one.

```
{%
  \ifstrempy{#3}{%
    \abb@abbrev@acc{#5}{#6#4}%
  }{%
    \abb@abbrev@wd@#3\@stop{#4}{#5}{#6}%
  }%
}%
}
```

Given that a candidate abbreviation has matched, determine whether the abbreviation should actually be performed, and execute it if so. An abbreviation should not be performed if the immediately following text is a word character, because that would result in breaking a word in the middle of an abbreviation.

#1 is the accumulated text, #2 is the abbreviation to add, #3 is the remaining text not including any of the text to be abbreviated, #4 is the command to run if we bail out (that is, if no abbreviation is to be made using this candidate), #5 is the class, and #6 is the callback. (#5–#6 are left on the stack.)

```
\def\abb@abbrev@found#1#2#3#4{%
  %
  \ifstartswithwordchar{#3}{%
    #4%
  }{%
    \ifstrempy{#2}{%
      \abb@abbrev@delete{#1\abb@abbrev@delete!#3}{#4}%
    }{%
      \find@try\find@start{%
        { }\abb@abbrev@maybeclop{ } }%
        { }\abb@abbrev@maybeclop{.} }%
      }{#3}{\abb@abbrev@acc}%
      {#3}{#1#2}%
    }%
  }%
}
```

Given that an abbreviation can be executed and will only delete text, determine whether the abbreviation is executable. A deletion can be executed only if the remaining text is fully surrounded by spaces (or the beginning/end of text); otherwise joining punctuation might be left stranded. If it is executed, reconcile the remaining spaces.

#1 is the accumulated text plus the remaining text, separated by \abb@abbrev@delete!. #2 is the bailout macro sequence (to be executed if this abbreviation fails), #3 is the class, and #4 is the callback. (#3–#4 are left on the stack.)

```
\def\abb@abbrev@delete#1#2{%
  \find@in{ \abb@abbrev@delete! }{#1}{%
    \abb@abbrev@delete@
  }{%
    \find@eq{ \abb@abbrev@delete! }{#1}{%
      \abb@abbrev@end{ }%
    }{%
      \find@end{ \abb@abbrev@delete! }{#1}{%
        \abb@abbrev@end
      }{%
        \find@start{ \abb@abbrev@delete! }{#1}{%
          \abb@abbrev@acc
        }{%
          \@firstoftwo{#2}%
        }{ }% Empty argument is the empty accumulated text
      }%
    }%
  }%
}
\make@find@in{ \abb@abbrev@delete! }
\make@find@start{ \abb@abbrev@delete! }
\make@find@end{ \abb@abbrev@delete! }
\make@find@eq{ \abb@abbrev@delete! }
```

Given that an abbreviation led to a deletion and the abbreviation was surrounded by two spaces, delete one space and continue the abbreviation process. #1 is the accumulated text with a trailing space deleted, #2 the remaining text with a leading space deleted, #3 is the class, and #4 is the callback. (#3–#4 are left on the stack.)

```
\def\abb@abbrev@delete@#1#2{%
  \abb@abbrev@acc{ #2}{#1}%
}
```

#1 is the character to test for potential removal at the end of the accumulated text, #2 is the remaining text with the character removed, #3 is the remaining text without the character removed, and #4 is the accumulated text. #5 is the class, #6 the callback (#5-#6 are left on the stack).

```
\def\abb@abbrev@maybechop#1#2#3#4{%
  \find@end{#1}{#4}{%
    \@firstoftwo{\abb@abbrev@acc{#2}{#4}}%
  }{\abb@abbrev@acc{#3}{#4}}%
}%
\make@find@start{.}
\make@find@start{ }
\make@find@end{.}
\make@find@end{ }
```

3 Removal of Spaces Between Dotted Initials

`\abb@initialdots` $\{\langle text \rangle\}\{\langle dot \rangle\}\{\langle callback \rangle\}$ processes a text to remove dots between single-letter initials. Given a text, it will convert sequences such as “A. A. Milne” to “A.A. Milne” and “F. 4th” to “F.4th”. $\langle text \rangle$ is the text to process and $\langle callback \rangle$ is the callback to run on the resulting output.

$\langle dot \rangle$ is the dot character to search for. According to some abbreviation schemes, it is desirable not to collapse spaces between single initials when the initials refer to different classes of words. The macro `\abb@dot` can be used in the place of the period character to indicate this differentiation between classes.

`\abb@dot` For example, the text “Northern University Law Journal” may be abbreviated to “N.U. L.J.” with only the first and third spaces collapsed, but not the second because institution words should not be joined with publication words. This can be implemented by abbreviating publication words with `\abb@dot` (shown here for an abbreviation class `journal`):

```
\abb@new{journal}
\abb@add{journal}{Northern}{N.}
\abb@add{journal}{University}{U.}
\abb@add{journal}{Law}{L\abb@dot}
\abb@add{journal}{Journal}{J\abb@dot}
```

Now when the text is abbreviated with `\abb@abbrev`, it produces:

```
N.□U.□L\abb@dot□J\abb@dot
```

The dot removal algorithm can now be run twice, once with a period as $\langle dot \rangle$ and once with `\abb@dot`, thereby producing the desired text.

This uses `\find@next` on the text with a state machine having states described below. Each state has macros for each of the possible find results from `\find@next`. Additionally, these items are always left on the token stack:

- the processed text
- the deferred text
- the dot character
- the final callback

Initial entrypoint. #1 is the text, #2 is the character to treat as a dot, and #3 is the callback for the result.

```
\def\abb@initialdots#1#2#3{%
  \abb@id@a{#1}%
  {}% Initially the processed text is empty.
  {}% Deferred text is empty.
  {#2}% The dot character.
  {#3}% At end of the macro stack for use by \abb@id@done
}
\def\abb@id@done#1#2#3#4{%
  #4{#1#2}%
}
```

a. Initial state.

- On letter, go to b.
- Otherwise go to a.

```
\def\abb@id@a#1{%
  \find@next{#1}\abb@id@backtoa\abb@id@backtoa\abb@id@a@letter\abb@id@done
}
\def\abb@id@backtoa#1#2#3#4{%
  \abb@id@a{#2}{#3#4#1}{}%
}
\let\abb@id@a@space\abb@id@backtoa
\let\abb@id@a@group\abb@id@backtoa
\def\abb@id@a@letter#1#2#3#4{%
  \ifletter{#1}{%
    \abb@id@b{#2}{#3#4#1}{}%
  }{%
    \ifdigit{#1}{%
      \abb@id@d{#2}{#3#4#1}{}%
    }{%
      \abb@id@a{#2}{#3#4#1}{}%
    }%
  }%
}
```

b. Seen a single letter.

- On dot, go to c.
- On letter, go to d.
- Otherwise go to a.

```
\def\abb@id@b#1{%
  \find@next{#1}\abb@id@b@space\abb@id@a@group\abb@id@b@letter\abb@id@done
}
\let\abb@id@b@space\abb@id@backtoa
\let\abb@id@b@group\abb@id@backtoa
\def\abb@id@b@letter#1#2#3#4#5{%
  \ifletter{#1}{%
    \abb@id@d{#2}{#3#4#1}{}%
  }{%
    \test\ifx#1#5\fi{%
      \abb@id@c{#2}{#3#4#1}{}%
    }{%
      \abb@id@a{#2}{#3#4#1}{}%
    }%
  }%
  {#5}% Put the dot character back on the stack
}
```

c. Seen a dot after b.

- On space, go to e.
- On letter, go to b.
- Otherwise, go to a.

```
\def\abb@id@c#1{%
  \find@next{#1}\abb@id@c@space\abb@id@c@group\abb@id@c@letter\abb@id@done
}
\def\abb@id@c@space#1#2#3#4{%
  \abb@id@e{#2}{#3#4}{#1}%
}
\let\abb@id@c@group\abb@id@backtoa
\def\abb@id@c@letter#1#2#3#4{%
  \ifletter{#1}{%
    \abb@id@b{#2}{#3#4#1}{}%
  }{%
  }
```

```

\@ifdigit{#1}{%
  \abb@id@d{#2}{#3#4#1}{}%
}%
\abb@id@a{#2}{#3#4#1}{}%
}%
}%
}

```

d. In a word

- On letter, go to d.
- Otherwise go to a.

```

\def\abb@id@d#1{%
  \find@next{#1}\abb@id@d@space\abb@id@d@group\abb@id@d@letter\abb@id@done
}
\let\abb@id@d@space\abb@id@backtoa
\let\abb@id@d@group\abb@id@backtoa
\def\abb@id@d@letter#1#2#3#4{%
  \@ifwordchar{#1}{%
    \abb@id@d{#2}{#3#4#1}{}%
  }{%
    \abb@id@a{#2}{#3#4#1}{}%
  }%
}%
}

```

e. Seen a space after c

- On letter, go to f.
- On number, collapse the space and go to d.
- Otherwise, go to a.

```

\def\abb@id@e#1{%
  \find@next{#1}\abb@id@e@space\abb@id@e@group\abb@id@e@letter\abb@id@done
}
\def\abb@id@e@space#1#2#3#4{%
  \abb@id@e{#2}{#3}{#1}%
}
\let\abb@id@e@group\abb@id@backtoa
\def\abb@id@e@letter#1#2#3#4{%
  \@ifletter{#1}{%
    \abb@id@f{#2}{#3}{#4#1}%
  }{%
    \@ifdigit{#1}{%
      \abb@id@d{#2}{#3#1}{}%
    }{%
      \abb@id@a{#2}{#3#4#1}{}%
    }%
  }%
}%
}

```

f. Seen a single letter after e

- On dot, collapse the space and go to c.
- On letter, go to d.
- Otherwise, go to a.

```

\def\abb@id@f#1{%
  \find@next{#1}\abb@id@f@space\abb@id@f@group\abb@id@f@letter\abb@id@done
}
\let\abb@id@f@space\abb@id@backtoa
\let\abb@id@f@group\abb@id@backtoa
\def\abb@id@f@letter#1#2#3#4#5{%
  \@ifwordchar{#1}{%
    \abb@id@d{#2}{#3#4#1}{}%
  }{%
    \@test@ifx#1#5\fi{%
      \abb@id@f@stripspace{#1}{#2}{#3}{#4}%
    }{%
      \abb@id@a{#2}{#3#4#1}{}%
    }%
  }%
  {#5}%
}
\def\abb@id@f@stripspace#1#2#3#4{%
  \abb@id@c{#2}{#3#4#1}{}%
}

```

A token for dots.

```

\def\abb@dot{.}

```


1 Options

`\ProcessOptions\relax`